

Batsman

$$20 = S$$
$$750 = S$$

Average Formula

$$Av = \frac{750}{20} = 37.5$$

Equitable Dist.

Total sum 100

$$\begin{array}{c} 20 \\ \uparrow \\ \overline{Sc} \\ 20 \\ 20 \\ 20 \\ 20 \\ 20 \\ \hline 100 \end{array}$$

D M S Daugh
100 200 25 75

Sum

$$Av = \frac{\text{Sum}}{\text{Total Values}} = \frac{400}{4}$$

Total members

$$= \frac{400}{4}$$

$$= 100 \text{ Rs.}$$

$$\frac{400}{4} = 100$$

100 100 100 100

100 Rs.

$$Av = \frac{\text{Sum}}{\text{Total}}$$

CareerRide.com



E.D.

$$Av = \frac{\text{Sum}^*}{\text{Total No.}}$$

$$\text{Sum} = Av \times \text{Total}$$

80%

1

1 2 3 4 5
300 300 300 300 300

$$S = 1500$$

Total Sum

$$1 + 2 + 3 + 4 + 5$$

150 100 50 200 1000

$$\frac{\text{Rs. } 1500.}{5} = 300$$

Sum

Ages

1	2	3
15	25	40

↓

80 years

Batsman

1	2	3	4	5	6	7
1	10	15	12	2	3	5

48 runs

CareerRide.com



QUESTION (1)

Knowing that Vijay's expenditure for first 3 days is Rs. 100, Rs. 125 and Rs. 85, what is his 4th day expenditure as his 4 days average expenditure Rs. 90?

$$Av = \frac{S}{T} = 4$$

$$Av = \frac{310 + ?}{4}$$

$$90 = \frac{310 + ?}{4}$$

$$\therefore 360 = 310 + ?$$
$$? = 50 \text{ Rs.}$$

1	2	3	4
100	+ 125	+ 85	+ ?
			0
310 + ?			= S
			0

CareerRide.com



QUESTION (2) -

What will be average price of all the goods bought, if Ajay buys 30 erasers for Rs. 3 each, 35 chocolates for Rs. 10 each and 25 clips at the rate of Rs. 4 each?

$$\begin{aligned} \checkmark \text{AV} &= \frac{\checkmark \Sigma}{\checkmark T} \rightarrow \\ \checkmark T &= 30 + 35 + 25 \\ &= \underline{90 \text{ goods.}} \\ \text{AV} &= \frac{540}{90} = \underline{\underline{6 \text{ Rs.}}} \end{aligned}$$

$$\begin{aligned} 1e &= \text{Rs. } 3 \\ 30e &= \underline{\text{Rs. } 90} \\ 1c &= \text{Rs. } 10 \\ 35c &= \underline{\text{Rs. } 350} \\ 1d &= \text{Rs. } 4 \\ 25d &= \underline{\text{Rs. } 100} \end{aligned} \quad \left. \begin{array}{l} \\ + \\ + \end{array} \right\} = 540 \text{ Rs.}$$

QUESTION (3) -

Of the 20 cycles sold by Ajay, average cost of 12 cycles is Rs. 18000. In total he earned Rs. 300000. What was the average cost of remaining cycles?

$$Av = \frac{S}{T} \rightarrow ?$$
$$S = Av \times T$$

18000

$$12 \text{ cycled} = 2,16,000$$

$$20 \text{ cycles} = 3,00,000$$

$$8 \text{ cycles} = 84,000 \text{ Rs.}$$

$$Av_8 = \frac{S_8}{T} = \frac{84000}{8}$$

$$\text{Rs. } 10,500$$

$$12 \text{ cycles}$$

$$Av = 18000$$

$$18 \times 12$$

$$S = 18000 \times 12$$

$$S = 2,16,000 \text{ Rs.}$$

$$\text{Total 3 Lakh.}$$

QUESTION (4) -

Without considering the salary of the boss, the average salary reduces by Rs. 1000. What will be salary of boss if average salary of 11 employees and the boss is Rs. 18000?

$$Av = \frac{S}{T}$$
$$S = Av \times T \quad \checkmark$$
$$S = 18000 \times 12$$
$$S = 2,16,000 \text{ Rs.} \quad \checkmark$$

Diagram: A circle containing '12' has an arrow pointing to a circle containing '11 + BOSS'.

$$Av_{12} = 18000$$

$$Av_{11} = 18000 - 1000$$

$$\hookrightarrow = 17000 \quad \checkmark$$

$$S = 17000 \times 11$$

$$S = \text{Rs. } 1,87,000$$

= 11 employees

$$\begin{array}{r} 2,16,000 \\ - 18,7000 \\ \hline \end{array}$$

29,000 Rs.

QUESTION (5) -

Average age of 5 people is 42 years. Another group has 8 people who have average age of 81 years. When both groups are mixed what is average age of all people? 66 yrs.

Total Sum

$$Av = \frac{S}{T} = ?$$

$\uparrow S$
 $\uparrow T$ ✓

$5 + 8 = 13$ people sum + sum
 $\rightarrow 5 + 8$

$$Av = \frac{858}{13} = 66 \text{ years}$$

$210 + 648$
 $(858 \text{ years}) \rightarrow S$

$$Av = \frac{S}{T}$$
$$S = Av \times T$$

$$Av_5 = \frac{S_5}{T}$$
$$42 = \frac{S_5}{5}$$

$$S_5 = 210 \text{ years}$$

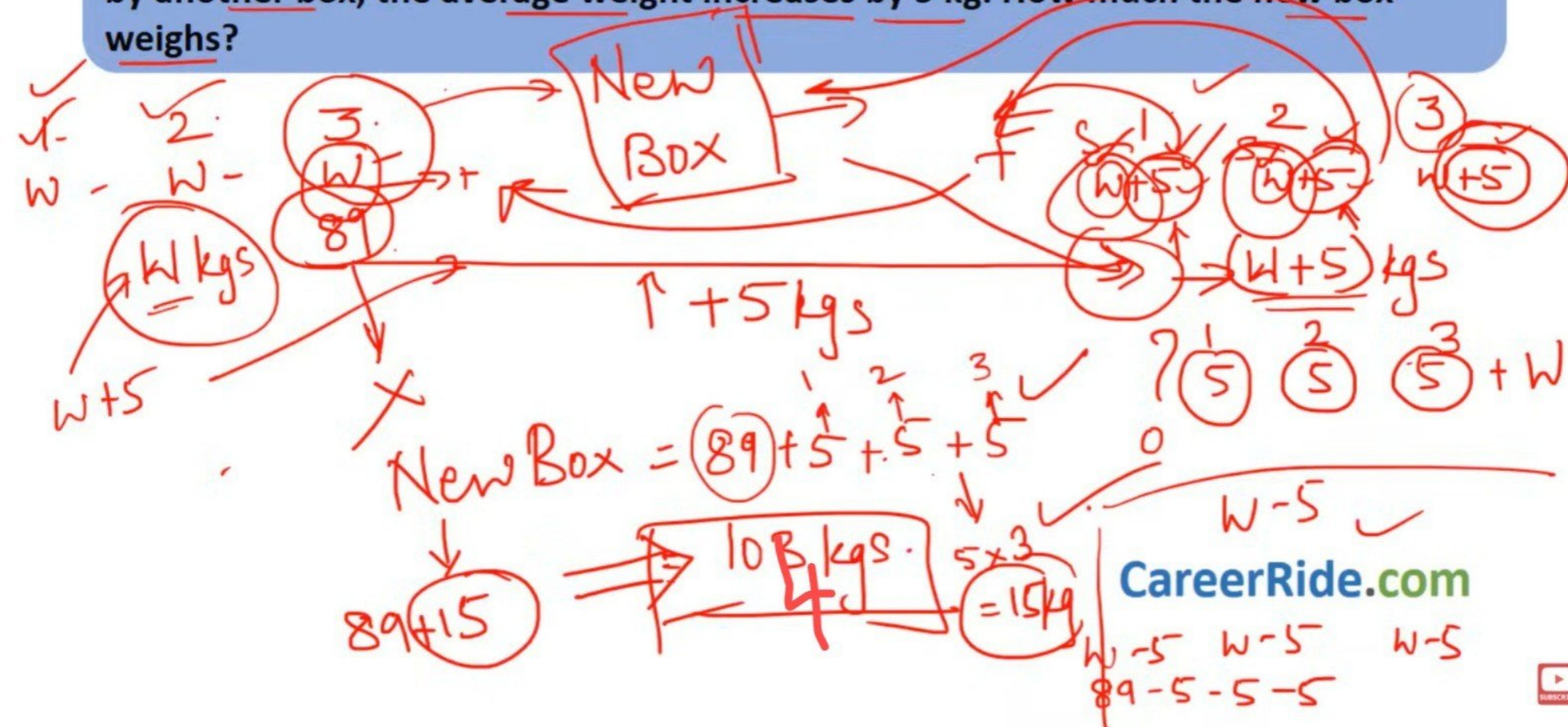
$$Av_8 = \frac{S_8}{T}$$

$$81 = \frac{S_8}{8}$$

$$S_8 = 648 \text{ years}$$

QUESTION (6) -

3 boxes have some average weight. When one box which weighs 89 kg is replaced by another box, the average weight increases by 5 kg. How much the new box weighs?



QUESTION (7) -

How old will Raju be if ratio of his age and one of his twin grandsons' is 11:2 and average age of his and his both grandsons is 50 years?

T.S.

$$Av = \frac{\sum}{T \rightarrow 3}$$

↓

$$50 = \frac{S_3}{3}$$

$$S_3 = 150 \text{ yrs} \checkmark$$

$$R_{AJV} = 11K$$
$$= 110 \text{ years}$$

$$R : G_1 = 11 : 2$$

$$R = 11K$$

$$G_1 = 2K = G_2$$

$$11K + 2K + 2K = 15K$$

$$15K = 150 \text{ yrs}$$

$$K = 10 \text{ yrs}$$

QUESTION (8) -

Had Ajay scored 18 runs more in his 3rd innings and 4 runs more in his 7th innings, his average would have become 66 runs. But it is 64 runs. How many innings did he play?

Handwritten solution:

N
64

$18 + 4 = 22 \text{ Runs more}$

$+2 \quad +2 \quad +2 \quad +2 \quad +2$

66

$+2$

$N \times 2 = 2N \text{ Runs}$

$2N = 22$
 $N = 11$

$Av = \frac{S}{T}$
 $S = Av \times T$
 $W+5 \quad W+5 \quad W+5$

QUESTION (9) -

In a group of people, the oldest and the youngest have an age difference of 100 years. If these two are left out of counting, then the average age of the remaining 40 people is 28. The average age of entire group being 30, how old is the eldest person?

'N' ✓ '100+N' ✓
Y ✓ O ✓

$$40 + Y + O = 42 \text{ people}$$

$$Av_{42} = 30 = \frac{S_{42}}{T}$$

$$30 = \frac{S_{42}}{42}$$

$$S_{42} = 1260 \text{ years}$$

$$Av_{40} = 28 = \frac{S_{40}}{T}$$

$$28 = \frac{S_{40}}{40}$$

$$S_{40} = 1120 \text{ years}$$

$$Y + O + (40)$$

$$(N) + (100 + N) + 1120 = 1260$$

$$2N = 1260 - 1120 = 140$$

$$2N = 140$$
$$N = 70$$

$$O = 100 + N = 170 \text{ yrs}$$

CareerRide.com

$$Av = \frac{S}{T}$$
$$S = Av \times T$$



QUESTION (10) -

A batsman played 11 innings and has a certain average. This average increases by 2 runs when his 3 innings of 32 runs, 33 runs and 34 runs are replaced by 3 other innings. Find the average of these 3 new innings?

$$\begin{array}{lcl} 11 \text{ Innings} & + 2 & 2 \times 11 = 22 \text{ Runs} \rightarrow \text{Increase} \\ & \text{Av} & \end{array}$$

$$\begin{aligned} \text{New innings} &= \text{Old innings} + \\ \downarrow & \\ 3 & \\ &= (32 + 33 + 34) + 22 \\ &= (121 \text{ runs}) \Rightarrow S_{3N} \end{aligned}$$

$$\begin{aligned} \text{Av} &= \frac{S}{I} \\ S &= \text{Av} \times I \\ \downarrow & \quad \downarrow \quad \downarrow \\ \text{Av}_{3N} &= \frac{S_{3N}}{I} = \frac{121}{3} \\ &= 40\frac{1}{3} \text{ runs} \end{aligned}$$